Fig. 3: Author's prototype showing temperature set at 65°C

enclose it in a suitable box. Fix LED1 and LED2 in front side of the box and extend the sensor to the water tank with suitable wires.

Water temperature setting is important. The pot is used to set the water temperature between 40 and 70°C. As the pot is rotated, it gives analogue input of 0 to 5V at pin A0. The internal 10-bit ADC of ATMega328 microcontroller converts this analogue voltage into a digital value of 0 to 1023. This 0 to 1023 range is mapped into corresponding temperature range of 40 to 70°C, which is shown on the display.

SET and ON push-button switches give logic input 0 to pins A1 and A2 (digital pins 14 and 15) when pressed. SET is pressed when the required water temperature is set through the pot. This sets the temperature limit till the water will be heated. Pressing ON pushbutton switches the heater on and starts heating water.

When ON button is pressed, logic 1 is given to pin A5 (digital pin 19). This switches on the relay through opto-coupler. As the relay is switched on, the heater coil gets 230V and starts heating water. Green LED is turned on by sending logic 1 to pin A4 (digital pin 18) to indicate the heater is on.

PARTS LIST

Semiconductors:	
IC1	- MCT2E opto-coupler
LED1, LED2	- 5mm LED
D1	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1-R3	- 330-ohm
R4	- 4.7-kilo-ohm
VR1	- 10-kilo-ohm pot
Miscellaneous:	
BOARD1	- Arduino Nano
DIS1	- 4-digit 7-segment display
S1, S2	- Push-to-on switch
RL1	- 5V, SPDT relay
PZ1	- Piezo buzzer
CON1, CON2	- 2-pin connector
SENSOR1	- DS18B20 sensor
	- Water heater (1000-watt)

The heating water's temperature is sensed by DS18B20 probe, which gives temperature reading to Arduino. Arduino shows this temperature on display.

When water reaches the set temperature, logic 0 is sent to the board's pin A3, the relay switches off, and thus the heater turns off. Simultaneously, red LED is turned on by sending logic 1 to pin A3 (digital pin 17) to indicate the heater is off. Buzzer pin D0 gets logic 1 for a while (2 to 3 seconds) to produce the beep sound to indicate that the water has been heated to the extent desired. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*


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